

Unit 1 Notes

Foundations of Geometry

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 1 Notes: Foundations of Geometry

Big goal

Geometry begins with exact language. A drawing can suggest a relationship, but definitions, notation, measurements, and theorems let you communicate why a statement is true. In this unit, master vocabulary, angle relationships, constructions, and coordinate tools: slope, distance, midpoint, and line equations.

1. Undefined terms and basic objects

The undefined terms of geometry are **point**, **line**, and **plane**. We do not define them using simpler geometric words; instead, we understand them by their properties.

Object	Notation	Meaning
Point	A	An exact location. It has no size.
Line	$\overleftrightarrow{AB}$ or line ℓ	Extends forever in two opposite directions through points A and B .
Segment	$\overline{AB}$	Part of a line with endpoints A and B . It has a measurable length, written AB .
Ray	$\overrightarrow{AB}$	Begins at endpoint A , passes through B , and continues forever.
Plane	plane P or plane ABC	A flat surface extending forever in all directions. Three noncollinear points determine exactly one plane.
Angle	$\angle ABC$	Two rays with common endpoint B . The middle letter is the vertex.

Notation precision

$\overline{AB}$ is the segment. AB is the length of the segment. $\overleftrightarrow{AB}$ is the line through A and B . A line has no length because it never ends.

2. Collinearity, coplanarity, and betweenness

Points are **collinear** if they lie on the same line. Points are **coplanar** if they lie in the same plane. A point B is **between** A and C when A, B, C are collinear and $AB + BC = AC$.

Segment Addition Postulate

If B is between A and C , then $AB + BC = AC$. Use this whenever a segment is split into smaller pieces.

A **midpoint** divides a segment into two congruent segments. If M is the midpoint of $\overline{AB}$, then $AM = MB$. A **segment bisector** is a point, line, ray, segment, or plane that intersects a segment at its midpoint. A line itself cannot be bisected, because it has infinite length and no midpoint.

3. Angles and angle measurement

An angle is measured in degrees. An angle can be classified as acute ($0^\circ < m < 90^\circ$), right (90°), obtuse ($90^\circ < m < 180^\circ$), straight (180°), or reflex ($180^\circ < m < 360^\circ$).

Angle Addition Postulate

If point D lies in the interior of $\angle ABC$, then

$$m\angle ABD + m\angle DBC = m\angle ABC.$$

Use this exactly like segment addition, but with angle measures.

4. Angle pairs and theorems

Two angles are **adjacent** if they share a vertex and a side, and their interiors do not overlap. A **linear pair** is a pair of adjacent angles whose noncommon sides form a straight line. Linear pairs are supplementary, so their measures add to 180° . **Vertical angles** are opposite angles formed by two intersecting lines. Vertical angles are congruent.

Relationship	Fact to remember
Complementary angles	Measures add to 90° . They do not have to be adjacent.
Supplementary angles	Measures add to 180° . They do not have to be adjacent.
Linear pair	Adjacent supplementary angles.
Vertical angles	Congruent opposite angles made by intersecting lines.
Perpendicular lines	Lines that intersect to form right angles. If one angle is 90° , all four are 90° .

5. Constructions

A formal construction uses only allowed tools such as a compass and straightedge. A straightedge draws lines; it does not measure. A compass copies distances and draws arcs; it is the tool that transfers equal lengths.

Core construction recipes

Copy a segment: Draw a ray, set the compass to the original segment length, and mark that distance on the ray.

Copy an angle: Draw a base ray. Draw the same arc through both sides of the original angle. Copy that arc to the new angle. Then copy the distance between the arc-intersection points.

Bisect a segment: From both endpoints, draw equal-radius arcs above and below the segment. Connect the arc intersections. This perpendicular bisector hits the segment at its midpoint.

Bisect an angle: Draw an arc crossing both sides of the angle. From those two points, draw equal arcs that intersect. Connect the vertex to that intersection.

Construct a perpendicular line: Make equal arcs on both sides of a point, then use equal-radius arcs to locate a line through the point at 90° .

Construct a parallel line through a point: Copy the angle made by a transversal so that corresponding angles are congruent.

6. Perpendicular bisectors and equidistance

The perpendicular bisector of $\overline{AB}$ is the line perpendicular to $\overline{AB}$ through its midpoint. Every point on the perpendicular bisector is equidistant from A and B . Conversely, if a point is equidistant from A and B , then it lies on the perpendicular bisector of $\overline{AB}$.

7. Coordinate geometry formulas

For points $A(x_1, y_1)$ and $B(x_2, y_2)$:

Coordinate formula bank

$$\text{Distance: } AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Midpoint: } M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{Slope: } m = \frac{y_2 - y_1}{x_2 - x_1} \quad (x_2 \neq x_1)$$

The distance formula is the Pythagorean Theorem on the coordinate plane. The horizontal change is $x_2 - x_1$, the vertical change is $y_2 - y_1$, and the segment is the hypotenuse.

8. Lines, slopes, and equations

Parallel nonvertical lines have equal slopes. Perpendicular nonvertical lines have slopes that are negative reciprocals, meaning their product is -1 . Horizontal lines have slope 0. Vertical lines have undefined slope. A horizontal line and a vertical line are perpendicular.

Line equation forms

Slope-intercept form: $y = mx + b$, where m is slope and b is the y -intercept.

Point-slope form: $y - y_1 = m(x - x_1)$, which is best when you know a point and a slope.

To write a parallel line, keep the same slope. To write a perpendicular line, use the negative reciprocal slope. Then substitute the given point.

9. Coordinate proofs and classification

A coordinate proof uses formulas to prove a geometric relationship. Common evidence:

- equal distances prove congruent segments;
- equal slopes prove parallel lines;
- negative reciprocal slopes prove perpendicular lines;
- equal diagonal midpoints prove the diagonals bisect each other;
- distance plus slope evidence can classify figures such as triangles, parallelograms, rectangles, rhombi, and squares.

10. Worked examples

Example 1: Segment addition

If B is between A and C , $AB = 2x + 5$, $BC = 3x - 1$, and $AC = 44$, then

$$(2x + 5) + (3x - 1) = 44 \Rightarrow 5x + 4 = 44 \Rightarrow x = 8.$$

So $AB = 21$, $BC = 23$, and $AC = 44$.

Example 2: Angle relationships

If two vertical angles measure $7x - 9$ and $4x + 30$, set them equal:

$$7x - 9 = 4x + 30 \Rightarrow 3x = 39 \Rightarrow x = 13.$$

Each angle is 82° .

Example 3: Parallel and perpendicular lines

A line parallel to $y = -2x + 6$ through $(3, 1)$ has slope -2 :

$$y - 1 = -2(x - 3) \Rightarrow y = -2x + 7.$$

A line perpendicular to it has slope $\frac{1}{2}$, because $-2 \cdot \frac{1}{2} = -1$.

Example 4: Classifying with coordinates

For $A(0, 0)$, $B(4, 0)$, $C(4, 3)$, slopes $AB = 0$ and BC is undefined, so $AB \perp BC$. Distances are $AB = 4$, $BC = 3$, and $AC = 5$, so the triangle is a right triangle.

Mastery checklist

You are ready for Unit 1 when you can define and use correct notation, solve segment and angle equations, explain construction steps, calculate distance/midpoint/slope, write parallel and perpendicular line equations, and classify coordinate figures using evidence instead of appearance.